

NFL Wide Receiver Efficiency Frontier

Cap Spending vs. On-Field Production — A Pittsburgh Steelers Case Study (2020–2025)

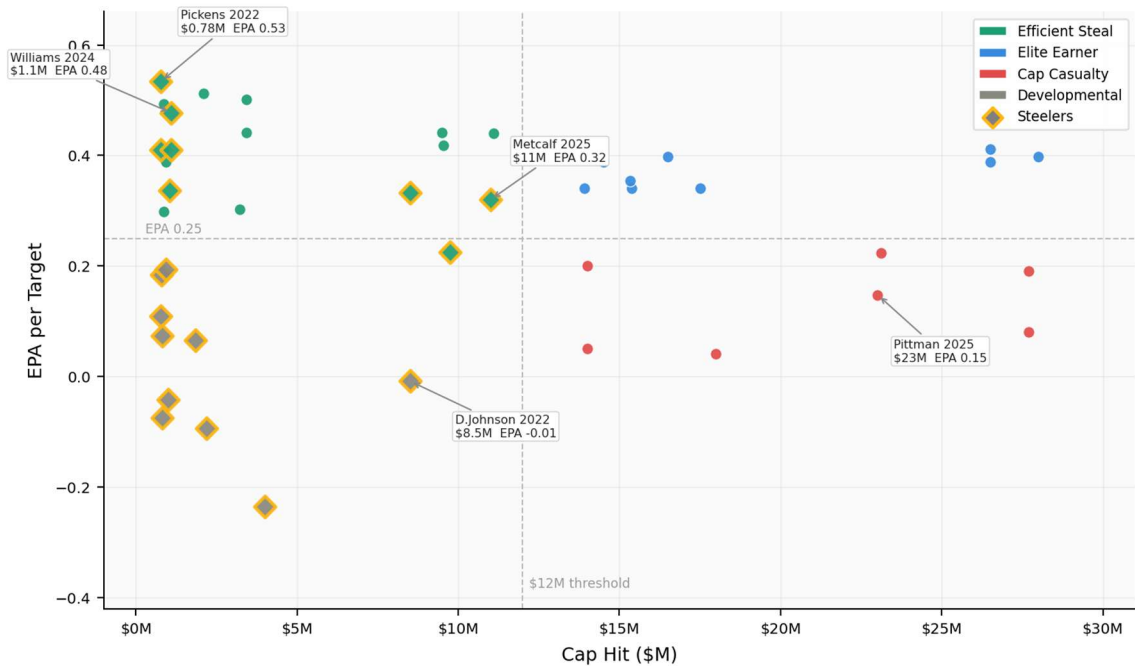


Figure 1. Each point is one player-season (2020–2025). Gold-outlined diamonds are Pittsburgh Steelers. Dashed lines mark the \$12M cap and 0.25 EPA/target thresholds separating the four efficiency quadrants.

Legend

Color / Shape	Meaning
Green	Efficient Steal — cap under \$12M, EPA above 0.25. The best value quadrant.
Blue	Elite Earner — high cap, high EPA. Expensive, but the production backs it up.
Red	Cap Casualty — high cap, low EPA. The player is costing more than he is producing.
Gray	Developmental — low cap, low EPA. Rookies, depth players, and injury-shortened seasons.
Gold outline	Pittsburgh Steelers player-season (diamond marker).
Dashed lines	Quadrant boundaries: \$12M cap hit (vertical) and 0.25 EPA/target (horizontal).

Supporting Figure: Steelers EPA Trajectory (2020–2025)

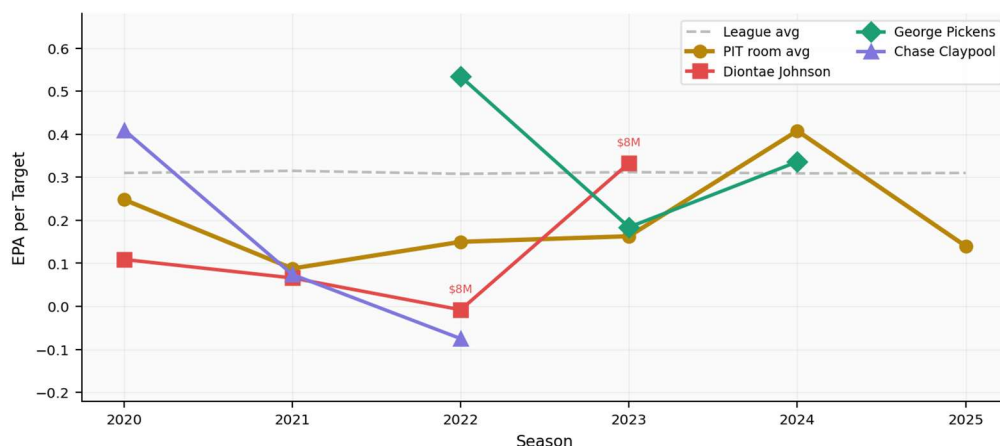


Figure 2. EPA per target over time for three Steelers receivers, alongside the Pittsburgh room average (gold) and league average (dashed gray). Dollar labels flag seasons following a contract extension.

Findings

- Pittsburgh averaged **0.179 EPA/target** over six seasons against a league average of **0.310**. That gap persists across every year in the dataset — it is not noise; it is a pattern.
- The most efficient Steelers WR season in the data was George Pickens in 2022: EPA of 0.534 on an \$0.78M rookie deal. He followed that with 0.336 in 2024 on a similarly cheap contract. Pittsburgh traded him to Dallas for a third-round pick. His 2025 season in Dallas: EPA of 0.640 — which would have been the highest individual mark in the entire six-year dataset.
- Figure 2 traces the Diontae Johnson arc. He was a reasonable value at 0.109 EPA in 2020. After signing his extension, his EPA fell to 0.066 in 2021 and then hit -0.008 in 2022 on an \$8.5M cap hit. Michael Pittman Jr. is running a nearly identical sequence right now: \$23M cap hit, EPA of 0.148 in 2025.
- DK Metcalf's 2025 line (EPA 0.320) is above league average. His reputation in Pittsburgh suffered, but the numbers are not the problem. The issue is a four-year, \$132M extension signed before a new quarterback had been identified and before he had played a game in the system.
- The 2024 WR room — Pickens, Austin, and Williams — produced three Efficient Steal seasons on roughly \$3.25M combined. Pittsburgh chose to replace that group with Metcalf and Pittman at approximately \$34M, a tenfold increase in cost that produced lower aggregate EPA.
- Every Steelers WR season that lands in the Efficient Steal quadrant traces back to a rookie or minimum contract. There are no exceptions in six years of data. Cincinnati's Chase-Higgins window (2021–2023) shows what that model looks like at scale: two first-round picks on rookie deals, elite production, and enough cap space left over to build around them.

Significance

The NFL salary cap is the same number for every team. What separates efficient franchises from ones that spin their wheels is not budget size but allocation timing. Wide receiver has become the most expensive position to build through free agency, with top contracts clearing \$30M per year, yet the data shows that production rarely scales with price at this position.

Pittsburgh offered a particularly sharp case study because the decisions are well-documented and the outcomes are unambiguous. A trade that looked defensible in the moment looks very different once you put the EPA numbers next to the cap numbers. An extension that aged badly almost immediately. A quietly excellent 2024 receiver room

that was dismantled before it had a chance to develop. These are not obscure findings — they are visible in the data if you look.

Using EPA per target rather than traditional receiving stats matters because EPA accounts for situational context. A receiver who catches nine yards on third-and-ten is more valuable than one who catches nine yards on second-and-one. Yards from garbage time look the same in the box score as yards in a two-minute drill. EPA does not treat them the same. That distinction is what makes the efficiency frontier a more honest picture of roster value than any leaderboard based on raw counting stats.

Data & Methods

Data Sources

Play-by-play EPA: nfl_data_py, an open-source Python wrapper for the nflfastR play-by-play database. Six full seasons (2020–2025) were imported, covering roughly 270,000 rows and 372 columns per season (approximately 1GB total). The data was filtered to passing plays with a credited receiver to compute EPA per target per player per season. 76 of 78 target player-seasons matched successfully; the two misses were Calvin Austin III seasons in which he had zero targets.

Cap hit data: Spotrac.com WR cap hit rankings, 2020–2025. Current season figures pulled from the live Spotrac rankings page; historical seasons pulled from Spotrac's archived year pages and cross-referenced against Over The Cap (overthecap.com).

Supporting grades: PFF receiving grades were referenced during exploratory analysis as a secondary check on cluster assignments. All final clustering and reported findings use EPA per target exclusively.

Contextual reporting: Transaction details and season narratives drawn from ESPN, Behind the Steel Curtain, and Steelers Depot.

Analytical Methods

- **EPA per target** was selected over traditional counting stats (yards, receptions, touchdowns) because it captures situational value. Every targeted pass, including incompletions, drops, and interceptions thrown by the quarterback following a route, counts against the receiver's average. This penalizes volume receivers who accumulate targets without creating real positive value.
- **The efficiency frontier** places cap hit on the x-axis and EPA per target on the y-axis, creating a two-dimensional space where the relationship between cost and production becomes visible without requiring a single composite index.
- **KMeans clustering (k=4)** was applied via scikit-learn to standardized features. The choice of k=4 was validated using the elbow method and silhouette score analysis across k=2 through k=8. Cluster labels (Efficient Steal, Elite Earner, Cap Casualty, Developmental) were assigned post-hoc based on centroid positions in original feature space.
- **Name harmonization:** nfl_data_py encodes player names in F.Lastname format. A 40-player manual crosswalk was built to join the cap dataset with the EPA dataset on combined [player, season] keys.
- **Trajectory figure:** Figure 2 was constructed separately to show EPA per target over time for individual players, making the contract extension arc visible in a way the scatter plot alone cannot convey. The Pittsburgh room average was computed as the mean EPA across all Steelers WRs with targets in a given season.

Code & Data Archive

All Python code, the cap dataset, and this report are archived at: <https://github.com/Se7enB2st/NFL-Efficiency-Frontier>. The repository includes the full Jupyter notebook, a CSV of cap hit data by season, and a README with environment setup instructions for reproducing the analysis.